

Q: 2(i)

A: 2(i)

speed: 3.5 km/h

time: 12 hrs

distance: $s = vt$

$$= 3.5 \times 12$$

$$= 42 \text{ km}$$

$$= 4.2 \times 10^1 \text{ km}$$

Q: 2(ii)

A: 2(ii)

* fahrenheit: $F = \frac{9}{5} \times -89.2^\circ\text{C} + 32$

$$= -160.56 + 32$$

* kelvin: $K = -89.2 + 273$

$$183.8 \text{ K}$$

Q: 2 (iii)
A: 2 (iii)

(a) Transitive property of equality.

(b) Multiplicative property of inequality.

Q: 2 (iv)
A: 2 (iv)

(a) 2.4324

Characteristic: 2

Mantissa: .4324

antilog: 2.4324 = 2704 + 3

antilog $_{2.4324} = 270.7$

(b) 1.5890

Characteristic: 1

Mantissa: .5890

= 3882 + 0

antilog $_{1.5890} = 38.82$

$$\text{① } \log_2(7-x), \quad x=0$$

$$\log_2(7-0)$$

$$\log_2(7) \quad (\text{defined})$$

$$\text{② } x=1$$

$$\log_2(7-1)$$

$$\log_2(6) \quad (\text{defined})$$

$$\text{③ } x=6$$

$$\log_2(7-6)$$

$$\log_2(1) \quad (\text{defined})$$

$$\text{④ } x=7$$

$$\log_2(7-7)$$

$$\log_2(0) \quad (\text{undefined})$$

Q. 2(vii)

A. 2(vii)

$$\begin{aligned} \textcircled{a} \quad & 27 + 54n^3 \\ & (3)^3 + (8n)^3 \\ & (3 + 8n) \left[(3)^2 - (3)(8n) + (8n)^2 \right] \\ & (3 + 8n) (9 - 24n + 64) \end{aligned}$$

$$\begin{aligned} & a^3 - 64 \\ & (a^3 - 8^2) \\ & (a^3 - 8)(a^3 + 8) \\ & (a^3 - 2^3)(a^3 + 2^3) \\ & (a - 2)(a^2 + (a)(2) + 2^2) (a + 2)(a^2 - (a)(2) + 2^2) \\ & (a - 2)(a^2 + 2a + 4) (a + 2)(a^2 - 2a + 4) \\ & (a - 2)(a + 2) (a^2 + 2a + 4) (a^2 - 2a + 4) \end{aligned}$$

Q. 2(viii)

A. 2(viii)

$$\textcircled{a} \quad 12n^2 + n - 1$$

$$12n^2 + 4n - 3n - 1$$

$$4n(3n + 1) - 1(3n + 1)$$

$$(4n - 1)(3n + 1) \checkmark$$

$$\begin{aligned}
 & \textcircled{b} \quad 15n^2 + 8n + 1 \\
 & \quad 15n^2 + 5n + 3n + 1 \\
 & \quad 5n(3n+1) + 1(3n+1) \\
 & \quad (5n+1)(3n+1) \checkmark
 \end{aligned}$$

$$\text{HCF} \therefore 3n+1$$

$$\text{Q: } 2 \text{ (in)}$$

$$\text{A: } 2 \text{ (in)}$$

$$25a^4 - 30a^3 + 9a^2$$

~~$$(5a)^2 (3)^2 - 2(5a)(3) + (3)^2$$~~

$$a^2(25a^2 - 30a + 9)$$

$$a^2((5a)^2 - 2(5a)(3) + (3)^2)$$

$$a^2(5a - 3)^2$$

$$+ a^2(5a - 3)$$

$$+ (5a^3 - 3a^2)$$

$$+ a^2(5a - 3)$$

$$+ [a(5a - 3)]^2$$

$$+ (5a^2 - 3a)^2$$

Q: 2(ni)
A: 2(ni)

Q: $5y + 12 > 0$
 $5y > -12 + 0$
 $5y > -12$
 $y > \frac{-12}{5}$

Q: $5y + 12 > 0$
 $5(3) - 12 > 0$
 $15 - 12 > 0$
 $3 > 0$

Yes, the
satisfies

Q: $5 - 2n > -4n + 5$
 $5 - 2(4) > -4(4) + 5$
 $5 - 8 > -16 + 5$
 $-3 > -11$

Yes, the
satisfies

Q: 2(ni)
A: 2(nii)

$|5n - 3| = |n + 7|$
 $5n - 3 = n + 7$, $5n - 3 = -n - 7$
 $5n - n = +7 + 3$, $5n + n = +3 - 7$
 $4n = +10$, $6n = -4$
 $n = +\frac{10}{4}$, $n = -\frac{4}{6}$
 $n = +\frac{5}{2}$, $n = -\frac{2}{3}$

$n = -\frac{2}{3}$

Q. 2 (iii)

A. 2 (iii)

$$A(1.4, -1.5) \quad , \quad B(2.6, 3.5)$$

$x_1 \quad y_1 \quad \quad \quad x_2 \quad y_2$

$$\frac{1.4 + 2.6}{2}$$

$$, \quad \frac{-1.5 + 3.5}{2}$$

$$\frac{4}{2} \quad , \quad \frac{2}{2}$$

$$2, 1$$

Long Question

Q: 3 (i)

A: 3 (i)

a) $a = -2\sqrt{3} \quad b = -5$
 $-2(1.732) \geq -5$
 $-3.46 \geq -5$

$a=b, a>b, a<b$ are all trichotomy property so it is verified that $-3.46 > -5$ is trichotomy.

②

$$\text{Ratio} = 1+2+3 = 6$$

$$\text{① Aabika : } \frac{1,000,000}{6} \times \cancel{6,000,000}$$

$$= 1,000,000$$

$$\text{② Bilal : } \frac{1,000,000}{6} \times \cancel{6,000,000}$$

$$2,000,000$$

$$\text{③ Farzal : } \frac{1,000,000}{6} \times \cancel{6,000,000}$$

$$3,000,000$$

Q: 3(ii)
A: 3(ii)

(a) $\log \frac{28}{45}$

$\log \frac{4 \times 7}{9 \times 5}$

$\log \frac{4 \times 7}{3^2 \times 5}$

$\log 4 + \log 7 - \log 3^2 + \log 5$

$\log 2^2 + \log 7 - 2 \log 3 + \log 5$

$2 \log 2 + 0.8451 - 2(0.4771) + 0.6990$

$2(0.3010) + 0.8451 - 0.9542 + 0.6990$

$0.602 + 0.8451 - 0.9542 + 0.6990$

$1.4471 - 0.2552 = 1.6532$

~~1.4471~~ - 0.2061

(b) $\log(7^{40})$

$40 \log 7$

$40(0.8451)$

~~34.004~~ 34.004

Characteristic : $34 + 1 = 35$ digit.

$$\text{Q: } 3(\text{iv})$$

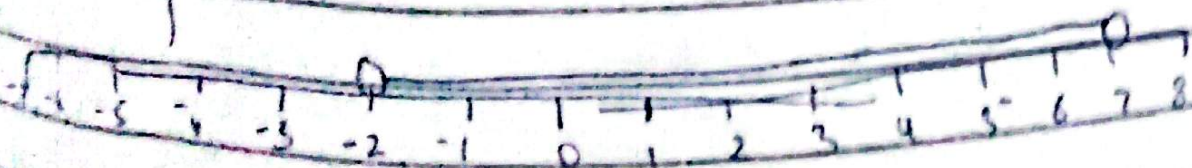
$$\text{A: } 3(\text{iv})$$

$$\begin{aligned} \textcircled{a} \quad 5x + 10 &< -35 & \text{or} & \quad 5x + 10 < -5 \\ 5x &< -35 - 10 & \text{or} & \quad 5x < -5 - 10 \\ 5x &< -45 & \text{or} & \quad 5x < -15 \\ x &< \frac{-45}{5} & \text{or} & \quad x < \frac{-15}{5} \\ x &< -9 & \text{or} & \quad x < -3 \\ & & & \quad x > -3 \end{aligned}$$

$$\{x/x \in \mathbb{R}, x < -9 \text{ or } x > -3\}$$

$$\textcircled{b} \quad |2y - 5| = 9$$

$$\begin{aligned} 2y - 5 &= 19 & \text{or} & \quad 2y - 5 = -9 \\ 2y &= 19 + 5 & \text{or} & \quad 2y = -9 + 5 \\ 2y &= 24 & \text{or} & \quad 2y = -4 \\ y &= \frac{24}{2} & \text{or} & \quad y = \frac{-4}{2} \\ y &= 12 & \text{or} & \quad y = -2 \\ \{12, -2\} & & & \end{aligned}$$



Q 3 (iv)

A = 3 (✓)

(a) i) Midpoint of E is AB :-

A(1, 0), B(5, 2)

$$\frac{1+5}{2}, \frac{0+2}{2}$$

$$\frac{6}{2}, \frac{2}{2}$$

$$3, 1$$

(ii) Midpoint of F is BC :-

B(5, 2), C(2, 6)

$$\frac{5+2}{2}, \frac{2+6}{2}$$

$$\frac{7}{2}, \frac{8}{2}$$

$$\frac{7}{2}, 4$$

iii) midpoint of G is CD :-

C(2,6) D(-3,2)

$$\frac{2+(-3)}{2}, \frac{6+2}{2}$$

$$\frac{2-3}{2}, \frac{8}{2}$$

$$\frac{-1}{2}, 4$$

iv) midpoint of H is DA :-

D(-3,2) A(1,0)

$$\frac{-3+1}{2}, \frac{2+0}{2}$$

$$\frac{-2}{2}, \frac{2}{2}$$

$$-1, 1$$

$$C(3, 1) \quad I\left(\frac{7}{2}, 3\right)$$

$$(b)(i) \quad dte = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{\left(\frac{7}{2} - 3\right)^2 + (3 - 1)^2}$$

$$= \sqrt{\left(\frac{7 - 3 \times 2}{2}\right)^2 + (3)^2}$$

$$= \sqrt{\left(\frac{7 - 6}{2}\right)^2 + 9}$$

$$= \sqrt{\left(\frac{1}{2}\right)^2 + 9}$$

$$= \sqrt{\frac{1}{4} + 9}$$

$$= \sqrt{\frac{1}{4} + \frac{9 \times 4}{4}}$$

$$= \sqrt{\frac{1 + 36}{4}}$$

$$= \sqrt{\frac{37}{4}} \Rightarrow \frac{\sqrt{37}}{2}$$

$$P\left(\frac{7}{2}, 4\right) \quad C_1\left(-\frac{1}{2}, 4\right)$$

$$ii) \quad d_{PG} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{\left(\frac{-1}{2} - \frac{7}{2}\right)^2 + (4 - 4)^2}$$

$$= \sqrt{\left(\frac{-8}{2}\right)^2 + (0)^2}$$

$$= \sqrt{\frac{64}{4} + 0}$$

$$= \sqrt{\frac{64}{4} + \frac{0^{xy}}{1^{xy}}}$$

$$= \sqrt{\frac{64 + 0}{4}}$$

$$= \sqrt{\frac{64}{4}}$$

$$= \frac{8^y}{x}$$

$$= 4$$

$$(ii) d_{CH} = \sqrt{\left(-1 - \left(-\frac{1}{2}\right)\right)^2 + (1 - 4)^2}$$

$$= \sqrt{\left(\frac{-1 + 1}{2}\right)^2 + (-3)^2}$$

$$= \sqrt{\left(\frac{-1^{\cancel{x2}} + 1}{\cancel{1^{\cancel{x2}}} 2}\right)^2 + (9)}$$

$$= \sqrt{\left(\frac{-2 + 1}{2}\right)^2 + (9)}$$

$$= \sqrt{\left(\frac{-1}{2}\right)^2 + 9}$$

$$= \sqrt{\frac{1}{4} + \frac{9 \times 4}{1 \times 4}}$$

$$= \sqrt{\frac{1 + 36}{4}}$$

$$= \sqrt{\frac{37}{4}}$$

$$= \sqrt{\frac{37}{2}}$$

$$H(-1, 1) \quad E(3, 1)$$

$$(iv) \quad dHE = \sqrt{(3 - (-1))^2 + (1 - 1)^2}$$

$$= \sqrt{(3 + 1)^2 + (0)^2}$$
$$= \sqrt{4^2 + 0}$$
$$= \sqrt{16}$$
$$= 4$$

1. D
2. B
3. B
4. C
5. D
6. A
7. A
8. A
9. A
10. B
11. D
12. C